Alexa Morales

Space Telescope Science Institute
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Education

University of Texas at Austin (UT Austin) — Graduated: Aug. 2026
College of Natural Sciences
Advisor: Steven Finkelstein
Ph.D. in Astronomy

University of Texas at Austin (UT Austin) — Aug. 2021 – Apr. 2024
College of Natural Sciences
Advisor: Steven Finkelstein
M.A. in Astronomy

Florida International University (FIU) — Aug. 2016 – Apr. 2021
College of Arts, Science, and Education
Bachelor of Science in Physics, Second Major in Natural and Applied Sciences, Minors in Mathematics & Astronomy

Technical Skills

Programming Languages: Python (primary), Bash, LaTeX, HTML
Scientific Python Ecosystem: NumPy, SciPy, Astropy, Matplotlib, emcee, Jupyter Notebook/JupyterLab, Conda, VSCode, Spyder
Development & Tools: Git/GitHub, HPC batch systems (SLURM), SAOImage DS9
Performance & Data Processing: Parallel batch processing on TACC super-computing systems (Stampede2/3, Lonestar6), memory optimization for large posterior sampling workflows
Operating Systems: macOS, Windows, Linux (HPC environments)

Research Experience

Postdoctoral Researcher at Space Telescope Science Institute — Sep. 2026 – Present
Advisor: Dr. Russell Ryan

- Implement Python-based data analysis pipelines for spectroscopic data for the *Roman Space Telescope* Supernova Project Infrastructure Team

Graduate Research Fellow at The University of Texas at Austin — Aug. 2021 – Aug. 2026
Observational Astrophysics – Advisor: Dr. Steven Finkelstein

- Designed and implemented Python-based data analysis pipelines for *HST* and *JWST* spectroscopic and photometric datasets
- Developed forward-modeling frameworks for UV spectral slope measurements using SED-fitting methodologies and generated scientific visualizations using Matplotlib and Astropy for large multi-wavelength datasets
- Built automated flux-calibration correction systems using Chebyshev polynomial fitting to reconcile spectroscopic and photometric measurements
- Implemented signal-to-noise filtering logic and conditional correction workflows to ensure physically consistent outputs
- Optimized posterior sampling workflows on TACC HPC systems; diagnosed and resolved memory allocation bottlenecks in batch modeling jobs
- Collaborated within large *JWST/HST* survey teams (UVCANDELS, CEERS, CAPERS) to validate data products and improve analysis methodologies

Research Assistant at Center for Astrophysics | Harvard & Smithsonian Jun. 2020 – Jul. 2021
Theoretical Astrophysics – Advisor: Dr. Charlotte Mason
- Developed numerical models in Python to simulate the redshift evolution of the Lyman-alpha luminosity function
- Integrated theoretical emission models with observed galaxy UV luminosity functions to generate predictive frameworks
- Performed parameter estimation and model validation across redshift bins using statistically robust fitting procedures

Research Assistant at Florida International University Jul. 2019 – Oct. 2019
Observational Astrophysics – Advisor: Dr. James Webb
- Studied the variability and activity of active galactic nuclei (AGN), i.e. galaxies centered with supermassive black holes, using optical telescopes
- Modeled microvariability of different objects, calculated their average absolute magnitudes, light curves, and their look-back in time with an average of 60 years' worth of data

Employment Experience

UT Astronomy Department: Teaching Assistant for Intro. to Astronomy Aug. 2022 – Dec. 2022
- Worked as an in-class tutor to help facilitate an active learning environment
- Held tutoring and review sessions outside of class to aid student understanding of various astronomy concepts

FIU Online: Course Developer Oct. 2019 – Aug. 2021
- Maintained, produced, tested, and quality assured courseware for online deployment
- Collaborated closely with a development team and faculty to design courses that met a technology-enriched distance education pedagogy

FIU Physics Department: Learning Assistant for Physics with Calculus II Jan. 2019 – May 2019
- Worked as an in-class tutor in order to help facilitate an active learning environment
- Aided student understanding of complex physics and math concepts through office hours and review sessions

Publications & Presentations

Papers in *ApJ* and *ApJL* (2021–Present)
- First Author (4) --165 citations, Total Publications (21) -- 2,700+ citations.
  See my full list of first and co-authored papers hyperlinked here.

Presented Talks & Posters
- **Seminar Speaker** – 'The Color of Origins: Tracing Early Galaxy Evolution Through Rest-Frame Ultraviolet Slopes with *HST* and *JWST*', Ph.D. Defense, UT Austin, TX, 2026
- **Invited Speaker (Public Outreach)** – 'What Can Colors Tell Us About Galaxies?', Sunset Park Elementary School, Miami, FL, 2025
- **Lightning Talk + Conference Poster** – 'Observed vs. Intrinsic UV Spectral Slopes for $z>4$ Galaxies with the *JWST* CAPERS Survey', CFC Conference, UT Austin, TX, 2025
- **Seminar Speaker** – 'Rest-Frame UV Spectral Slope Best Practices in the Era of JWST', UT Astronomy Galaxies and Cosmology Seminar, UT Austin, TX, 2025
- **Invited Speaker** – Université de Montréal Ciela Institute's Astromerique Student Talk Series: 'The Evolution of Rest-Frame UV Spectral Slopes with the UVCANDELS + NGDEEP Surveys at $z=2$-4 and $z=9$-16', 2024
- **Conference Poster** – 'Rest-Frame UV Colors for Faint Galaxies at $z \geq 9$ with the *JWST* NGDEEP Survey', First Stars Conference, Flatiron Institute, NYC, 2024
- **Invited Speaker** – STScI/JHU Exgal Seminar: 'Rest-Frame UV Colors for Faint Galaxies at $z \geq 9$ with the *JWST* NGDEEP Survey', 2024
- **Seminar Speaker** – 'Rest-Frame UV Colors for Faint Galaxies at $z \geq 9$ with the *JWST* NGDEEP Survey', UT Astronomy Galaxies and Cosmology Seminar, UT Austin, TX, 2024

- **Conference Poster** – ‘Rest-Frame UV Colors for Faint Galaxies at z ≥ 9 with the *JWST* NGDEEP Survey’, First Light Conference, MIT, MA, 2023
- **Seminar Speaker** – ‘The Evolution of Galaxy Rest-Frame UV Colors from z ~ 2-4 with HST UVCANDELS’, UT Astronomy Galaxies and Cosmology Seminar, UT Austin, TX, 2023
- **Invited Speaker** – UVCANDELS Special Session: ‘The Evolution of Galaxy Rest-Frame UV Colors in the GOODSN Field at z=2-4 with UVCANDELS’, AAS 241st Meeting, 2023
- **Seminar Speaker** – ‘The Evolution of the Lyman-Alpha Luminosity Function During Reionization’, UT Astronomy Galaxies and Cosmology Seminar, UT Austin, TX, 2022
- **Conference Speaker** – ‘The Evolution of the Lyman-Alpha Luminosity Function During Reionization’, SAZERAC Summer Conference, 2021
- **Conference Poster** – ‘The Evolution of the Lyman-Alpha Luminosity Function During Reionization’, AAS 237th Meeting, 2021

Academic Achievements & Awards

NSF Graduate Research Fellow	Sep. 2023 – Aug. 2026
AAS 237th Meeting Chambliss Astronomy Achievement Award (Honorable Mention)	Feb. 2021
NSF S-STEM Scholarship	Jan. 2018 – Apr. 2021
Inducted Member of Sigma Pi Sigma Physics National Honor Society	Apr. 2018
FIU Dean’s List	Aug. 2016 – Apr. 2021

Affiliations & Involvement

Member of the following *HST/JWST* collaborations: Aug. 2021 – Present

- UVCANDELS, CEERS, NGDEEP, COSMOS-Web, MEOW, THRILS, CAPERS

Referee for *The Astrophysical Journal*

Astronomy on Tap Austin **(Public Outreach)** Nov. 2023 – Aug. 2026

UT Austin Astronomy Vertically Integrated Projects (VIP) Program **(Mentor)** Aug. 2021 – Aug. 2026

Organizations:

UT Austin Astronomy E&I Organization	Aug. 2021 – Aug. 2026
FIU Sigma Pi Sigma Physics National Honor Society	Aug. 2016 – Apr. 2021
FIU AWSTEM (Society for the Advancement of Women in STEM)	Aug. 2017 – Apr. 2021
FIU Society of Physics Students	Aug. 2016 – Apr. 2021
American Physical Society	Aug. 2016 – Apr. 2021
FIU Astronomy Club	Aug. 2016 – Apr. 2021

Leadership Activities

FIU Society of Physics Students (SPS) Aug. 2018 – Apr. 2021

- Executive Board Member – CSO Representative

FIU Society for the Advancement of Women in STEM (AWSTEM) Aug. 2018 – Dec. 2019

- Executive Board Member – Secretary

Additional Information

Languages: Fluent in English & Spanish

Additional Relevant Courses: Regression Analysis, Survey of the Interstellar Medium, Computational Astrophysics, Gravitational Dynamics, Math Methods in Astrophysics, Radiative Processes & Radiative Transfer, Elements of Cosmology, Astronomical Data Analysis, Observational Astronomy + Lab, Modern Astrophysics, Mathematical Methods for Theoretical Physics, Ordinary and Advanced Partial Differential Equations, Linear Algebra, Calculus I-II-III, Statistical Methods I